Naive Bayes & Conditional Independence

Bayes Classifier & Independence Assumption

- Posterior rule: P(y | X) proportional to P(y) * Product_{i=1..n} P(x_i | y).
- Assumes all features x_i are conditionally independent given target class y.
- Reduces parameter estimation from exponential joint tables to O(n * |C|) likelihoods.

Log-Space MAP Decision Rule

- Predicting argmax_y [log P(y) + Sum log P(x_i | y)] prevents numerical underflow.
- Fast linear-time training and inference, well-suited for high-dimensional text.

Zero-Frequency Problem & Laplace Smoothing

Add-1 (Laplace) Smoothing Equation

- Unseen words produce $P(x_i \mid y) = 0$ and wipe out the entire class posterior.
- Laplace formula: $P(x_i \mid y) = (N_{y,i} + 1) / (N_y + |V|)$ where $|V|$ is vocabulary size.

Worked Example ($|V| = 1000$, $N_{Spam} = 200$)

- Seen token 'prize' ($N = 5$): $(5 + 1) / (200 + 1000) = 6 / 1200 = 0.005$.
- Unseen token 'urgent' ($N = 0$): $(0 + 1) / (200 + 1000) = 1 / 1200 = 0.000833$ (safe).